DBLP Dataset Analysis

Nokia Data Engineer – Summer Trainee Recruitment Task

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Repository:
github.com/zofiawarzycha/Nokia_DBLP_dataset_analysis

Abstract

This report presents a comprehensive analysis of the DBLP bibliographic dataset – a curated collection of over 6 million computer science publications maintained by Schloss Dagstuhl. Due to memory and processing constraints, the analysis is conducted on a subset of approximately **961 027 records** (up to the year 2024), covering conference papers, book chapters, books, and proceedings. The work encompasses six analytical dimensions: exploratory data analysis across publication volume, authorship, page count, venue, and title keywords; unsupervised topic discovery using both TF-IDF and sentence embeddings combined with KMeans clustering; a dense semantic search engine for paper retrieval; a Retrieval-Augmented Generation (RAG) pipeline powered by a local large language model; and a supervised publication-type classifier. Key findings include a near-exponential growth in CS publications since 2005, a measurable rise in collaborative authorship, and clear evidence of an LLM research explosion post-2021. Sentence embeddings significantly outperform TF-IDF for short-text clustering, producing visually separable topic regions in UMAP projections.

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1 Introduction and Dataset Overview

DBLP (*Digital Bibliography & Library Project*) is one of the most comprehensive open-access bibliographic databases in computer science. It is maintained by Schloss Dagstuhl – Leibniz Center for Informatics and contains metadata for over 6 million records including journal articles, conference papers, books, theses, and proceedings.

The dataset is distributed as a single XML file (`dblp.xml`) accompanied by a DTD schema file. The XML defines ten top-level record types, each with a different set of child elements. Rather than relying on external documentation, the schema was explored *empirically* in the first section of the analysis by streaming sample records from the file.

1.1 Analytical Goals

The primary goal is to extract as much insight as possible from the dataset. The analysis is structured around six technical pipelines, summarised in Table 1.

Table 1: Overview of analytical pipelines

#	**Pipeline**
1	Schema exploration via streaming XML parse
2	Targeted final parse of 1 M records into Parquet
3	Exploratory Data Analysis (5 dimensions)
4	Topic discovery: TF-IDF + KMeans vs. Sentence Embeddings + KMeans
5	Semantic search / paper recommendation engine
6	RAG pipeline: natural-language Q&A grounded in retrieved papers
7	Publication type classification (supervised ML)

1.2 Technical Environment

All code is implemented in Python 3.11 and executed in a Jupyter Notebook. Large intermediate files (Parquet datasets, embedding arrays) are cached to disk to avoid recomputation. The full dependency list is provided in `requirements.txt`. Key libraries include `pandas`, `lxml`, `scikit-learn`, `sentence-transformers`, `umap-learn`, `plotly`, and `requests` (Ollama API). The RAG pipeline requires Ollama with the `llama3.2` model running locally.

2 Data Parsing and Schema Exploration

2.1 XML Schema

The DBLP XML file is too large to load entirely into memory (uncompressed size exceeding 4 GB). All parsing is therefore implemented as a *streaming pass* using `lxml.etree.iterparse`, which processes one element at a time and immediately frees memory after extraction (`elem.clear()`).

A structural scan identified the ten top-level record types and their child fields. The results were cached to `dblp_sample_cache.json`. The most important record types and their key fields are:

- **inproceedings** – conference papers: *title, author, year, booktitle, pages, url*
- **article** – journal articles: *title, author, year, journal, volume, month*
- **proceedings** – conference proceedings: *title, editor, year, booktitle, isbn*
- **incollection** – book chapters: *title, author, year, booktitle, pages*
- **phdthesis** / **mastersthesis** – academic theses: *title, author, year, school*

A critical detail: the `.dtd` file must be passed to the parser with `load_dtd=True, resolve_entities=True` to correctly resolve HTML entities in author names (e.g. `ä` → *ä*). Without this, non-ASCII names from international authors appear corrupted.

2.2 Initial Parse – Data Quality Assessment

An initial broad parse loaded all ten record types up to 4 000 000 records. Table 2 shows the missing value rates for selected fields.

Table 2: Missing value rates – initial 4 M record parse (selected fields)

Field	Missing (count)	Missing (%)
address	3 999 997	100.00
chapter	3 999 998	100.00
month	3 998 441	99.96
school	3 996 528	99.91
journal	3 985 069	99.63
volume	3 952 138	98.80
publisher	3 918 741	97.97
pages	286 852	7.17
ee	153 107	3.83
author	72 435	1.81
title	1 792	0.04
year	0	0.00
record_type	0	0.00

The high sparsity of fields such as *isbn*, *series*, and *publisher* is expected: they are only meaningful for a subset of record types. The core analytical fields (*title*, *year*) are nearly complete.

2.3 Final Parse – Dataset for Analysis

Based on the quality assessment, the final parse was narrowed to five record types (*inproceedings*, *proceedings*, *incollection*, *book*, *article*) and six fields (*title*, *authors*, *year*, *venue*, *pages*, *record_type*). The record limit was set to 1 000 000 and records from 2025 onwards were excluded due to suspected incomplete indexing.

The resulting dataset contains **961 027 records**. Table 3 shows the distribution by record type.

Table 3: Record type distribution – final dataset

Record Type	Count
inproceedings	884 139
incollection	41 677
book	20 582
proceedings	14 617
article	12

Note: The very low article count (12 records) results from the 1 M parse cap: the DBLP XML file lists *mastersthesis* and *inproceedings* records first; journal articles appear later in the file and are therefore barely represented in the subset.

3 Exploratory Data Analysis

The EDA is structured across five analytical dimensions: publication volume over time, author behaviour, page count distributions, venue analysis, and title keyword trends.

3.1 Publication Volume Over Time

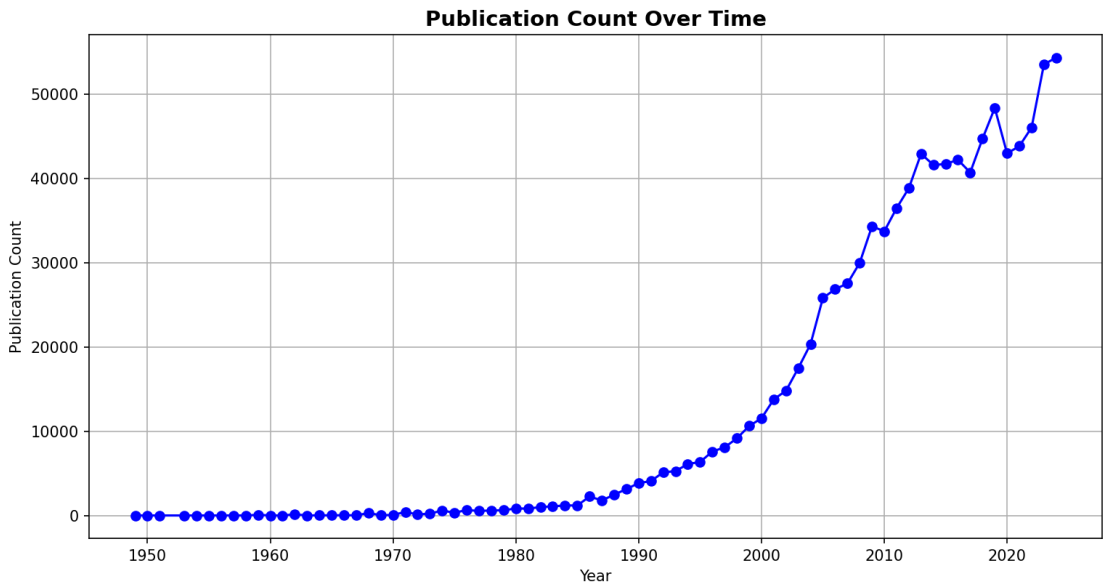

Figure 1: Annual publication count in DBLP (1950–2024).

Figure 1 shows a clear exponential growth in CS publications since the mid-1990s. Table 4 lists the ten most productive years.

Table 4: Ten most productive publication years

Year	Records
2024	54 335
2023	53 519
2019	48 359
2022	46 035
2018	44 704
2021	43 824
2020	42 965
2013	42 925
2016	42 201
2015	41 713

Growth accelerated sharply from 2005 onwards, coinciding with the rise of the internet, open-access publishing, and the global proliferation of CS conferences. The 2020 dip (-11.15%) likely reflects COVID-19 conference cancellations, followed by a strong rebound in 2021.

3.2 Author Analysis

3.2.1 Most Prolific Authors

Author names are decoded from HTML entities using Python's `html.unescape()` and DBLP's disambiguation suffixes (e.g. *Wei Wang 0002*) are intentionally preserved: removing them merges distinct researchers into a single identity. Table 5 lists the ten most prolific authors.

Table 5: Top 10 most prolific authors by publication count

Author	Papers	Debut	Last	Career (yrs)
H. Vincent Poor	385	1994	2024	30
Karl Henrik Johansson	328	2000	2024	24
Wen Gao 0001	326	1995	2024	29
Thomas S. Huang	316	1977	2020	43
Luca Benini	315	1996	2024	28
Miroslav Krstic	311	2000	2024	24
Rolf Drechsler	281	1992	2024	32
Hermann Ney	277	1978	2024	46
Aggelos K. Katsaggelos	269	1983	2024	41
Tamer Basar	268	2000	2024	24

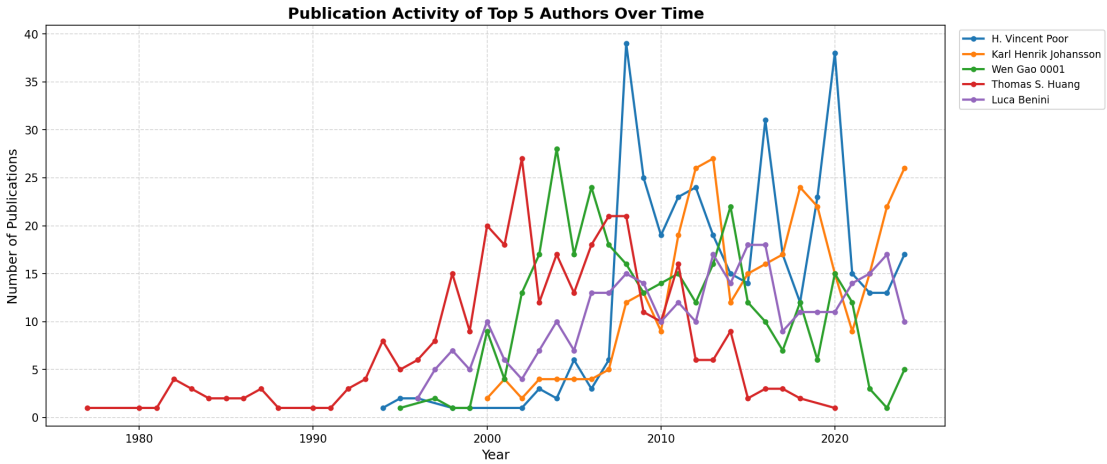

Figure 2: Annual publication activity of the five most prolific authors.

3.2.2 Team Size Trends

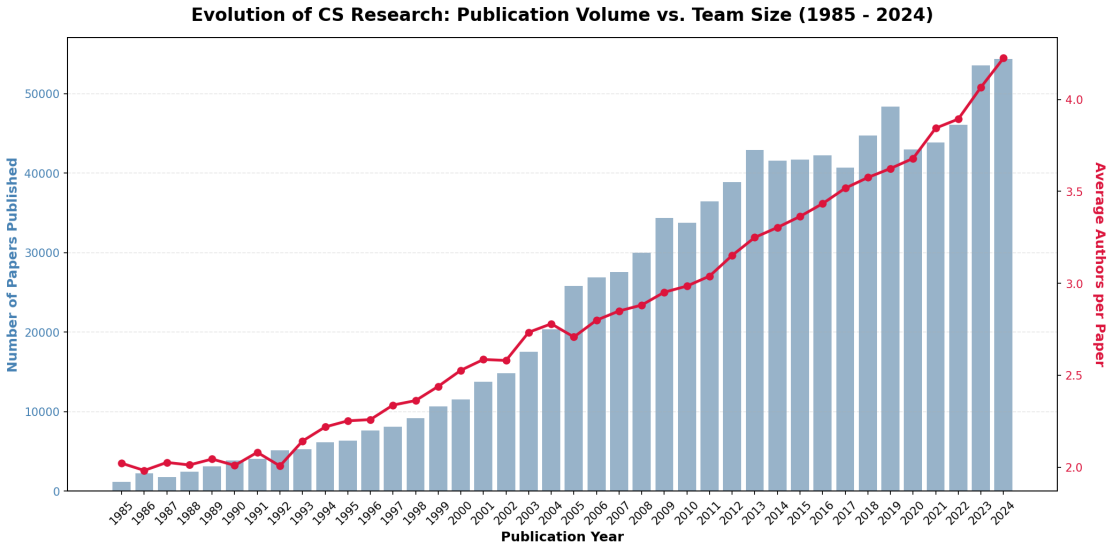

Figure 3: Publication volume (bars) and average authors per paper (line) 1985–2024.

Figure 3 demonstrates that the average number of authors per paper has grown from approximately 2.0 in 1985 to over 3.5 in 2024. This reflects a fundamental shift toward collaborative research in computer science, driven by the increasing complexity of research problems and international collaboration networks.

3.2.3 Collaboration Network and Top Pairs

All possible author pairs within each paper are generated using `itertools.combinations`. Table 6 lists the five most frequent co-authoring pairs.

Table 6: Top co-authoring pairs (joint paper count)

Author 1	Author 2	Joint Papers
Miki Haseyama	Takahiro Ogawa 0001	122
Olga Kosheleva	Vladik Kreinovich	103
David T. Blaauw	Dennis Sylvester	90
Shiguang Shan	Xilin Chen 0001	89
Dong-Seong Kim 0002	Jae-Min Lee 0001	86

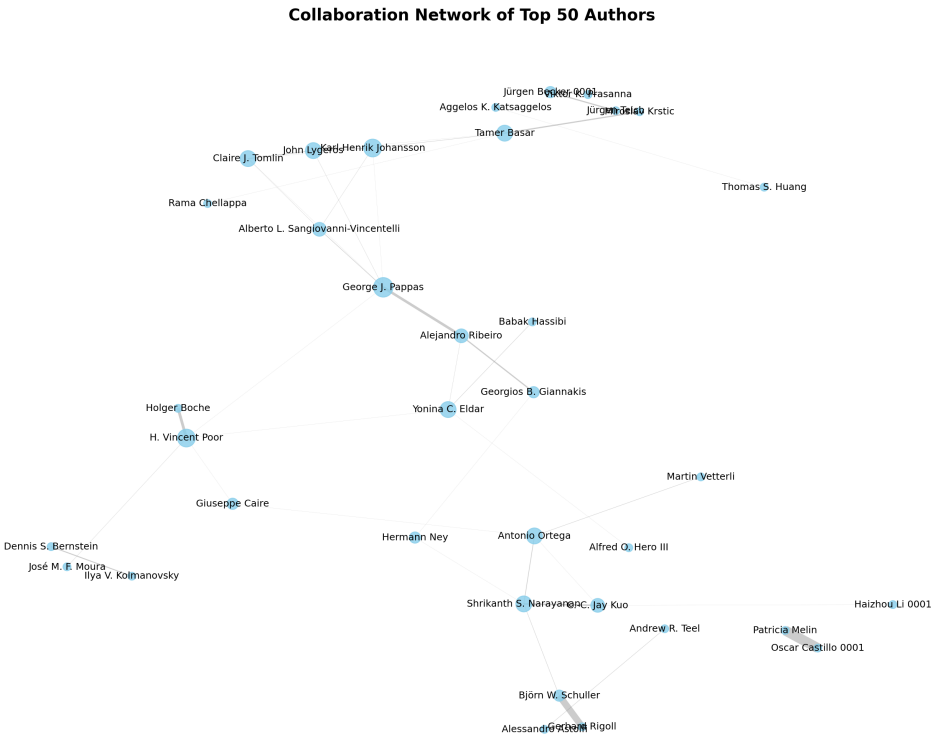

Figure 4: Collaboration network of the top 50 authors. Node size reflects degree; edge thickness reflects joint paper count.

The collaboration graph (Figure 4) reveals tight research clusters – for example, the control-systems group centred around Johansson and Krstic – as well as "bridge" researchers who connect otherwise disjoint communities.

3.3 Page Count Analysis

The raw `pages` field in DBLP is highly inconsistent, using formats such as `10-20`, `10-20`, `12:1-12:8`, and mixed Roman–Arabic combinations. A custom parser normalises all observed formats to a single integer page count; 895 329 publications were successfully parsed.

Table 7: Page count summary statistics (all parsed records)

Statistic	**Value**
Count	895 329
Mean	13.0
Median	6.0
25th pct	5.0
75th pct	10.0
Max	994

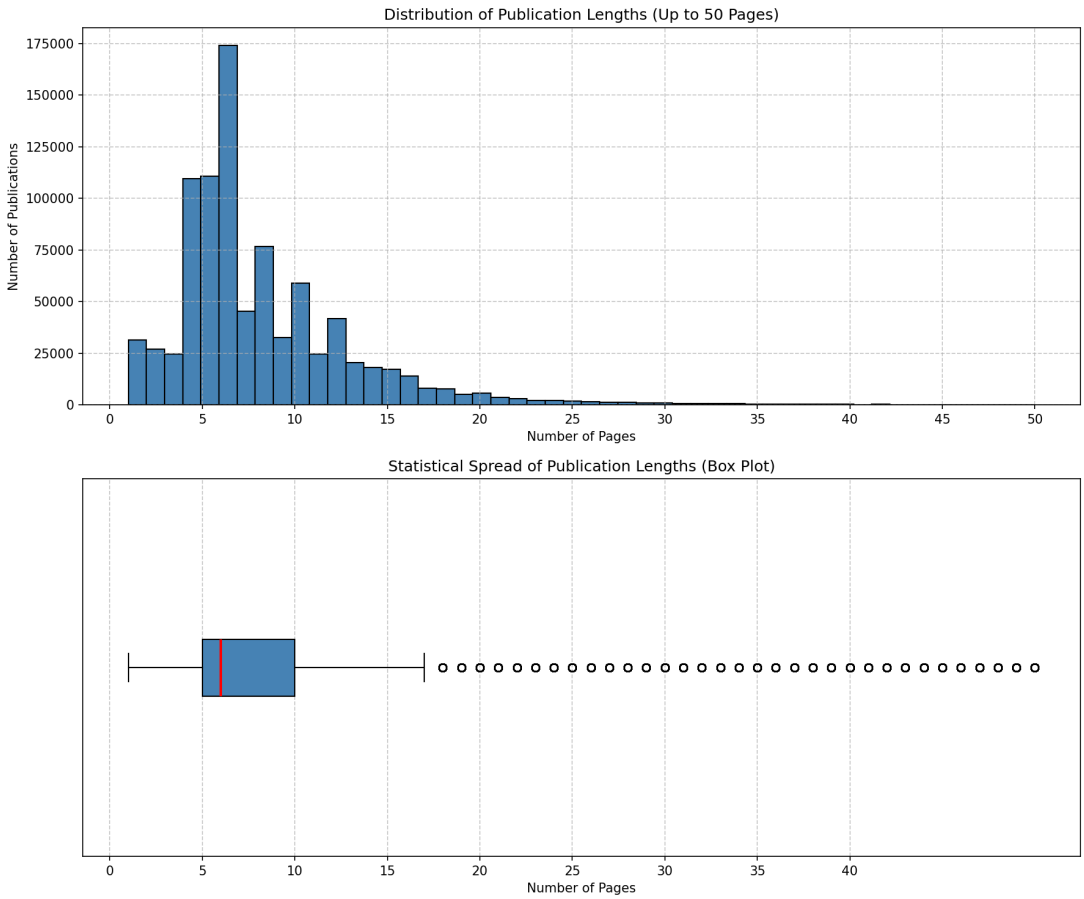

Figure 5: Distribution of publication lengths (up to 50 pages): histogram and box plot.

The bimodal distribution visible in Figure 5 reflects two dominant formats: short conference papers (4–8 pages) and longer book chapters or proceedings (10–20+ pages). The large mean–median gap (13 vs. 6) confirms strong right-skewness driven by a small number of very long publications (books/theses).

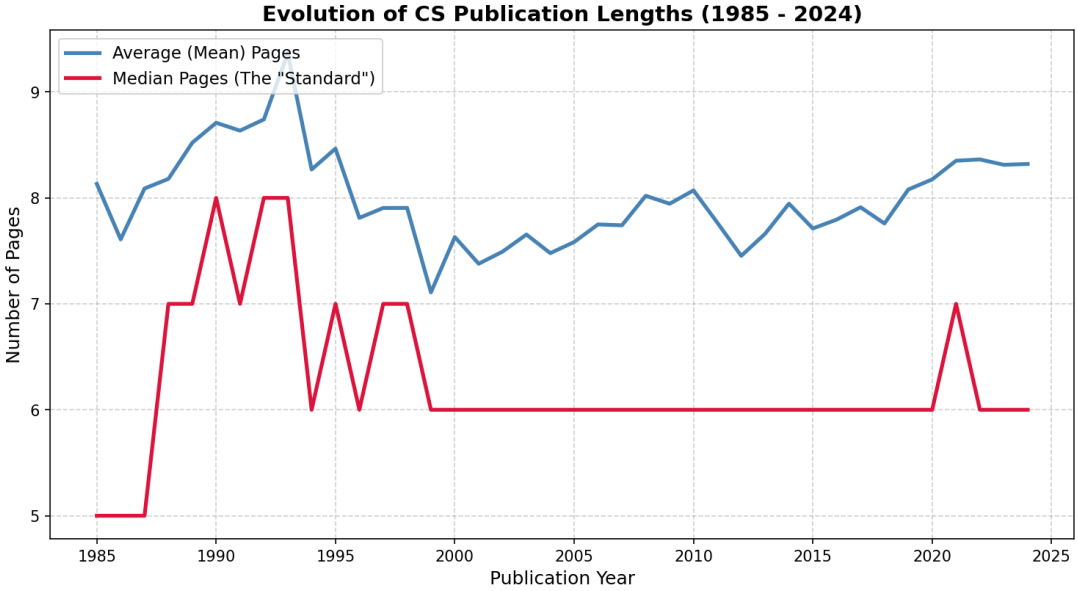

Figure 6: Mean and median paper length 1985–2024 (filtered to ≤50 pages).

Figure 6 shows that the median page length has remained remarkably stable at approximately 6 pages since 2000, despite a 5× increase in publication volume over the same period. This indicates that conference paper format norms have not eroded under growth pressure.

3.4 Venue Analysis

Table 8: Top 20 venues by paper count and average page length

Venue	Papers	Avg Pages
ICASSP	44 341	4.49
CDC	24 851	6.19
ACC	22 223	6.35
ICIP	17 977	4.69
ISIT	12 260	4.99
ECC	8 314	6.12
ICME	8 066	5.08
DATE	7 850	5.29
OFC	7 347	3.64
NIPS	7 234	8.51
NeurIPS	4 771	12.04
SODA	4 478	13.33
ACL (1)	4 094	14.53
IEEE BigData	5 331	7.41
IPDPS	3 856	7.80

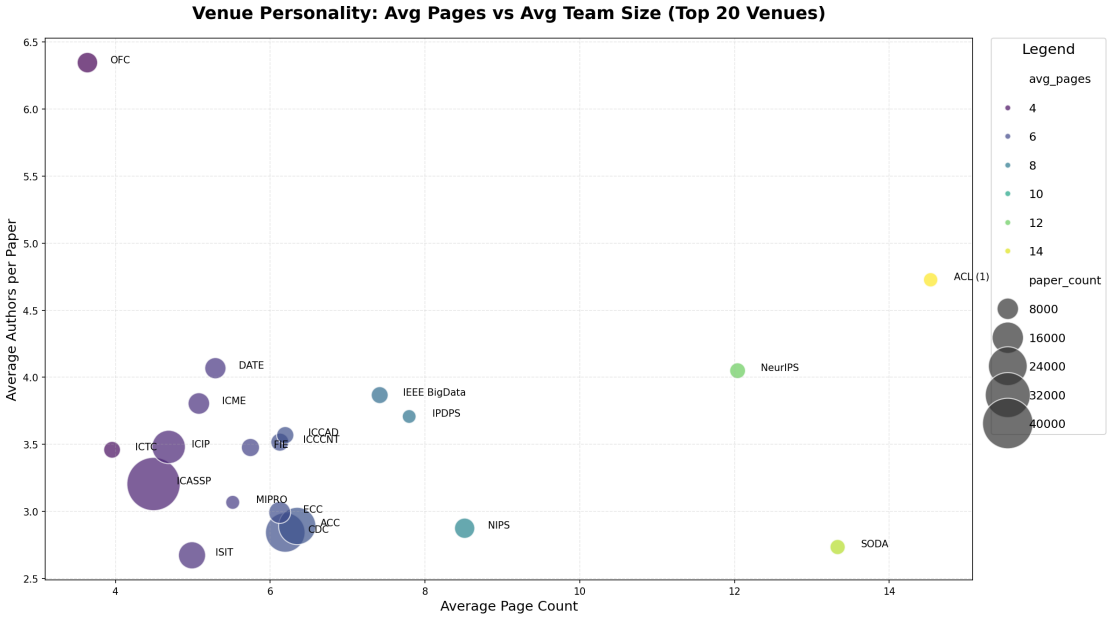

Figure 7: Venue personality: average page count vs. average team size for top 20 venues. Bubble area proportional to total paper count.

Figure 7 reveals distinct "personality profiles" for different venues. Signal-processing conferences (OFC, ICASSP, ICIP) cluster at the low-page-count end with moderate team sizes, while theory-oriented venues such as SODA and ACL feature longer papers. NeurIPS occupies a distinctive position with above-average page counts and a large author team size, reflecting the multidisciplinary nature of modern deep-learning research.

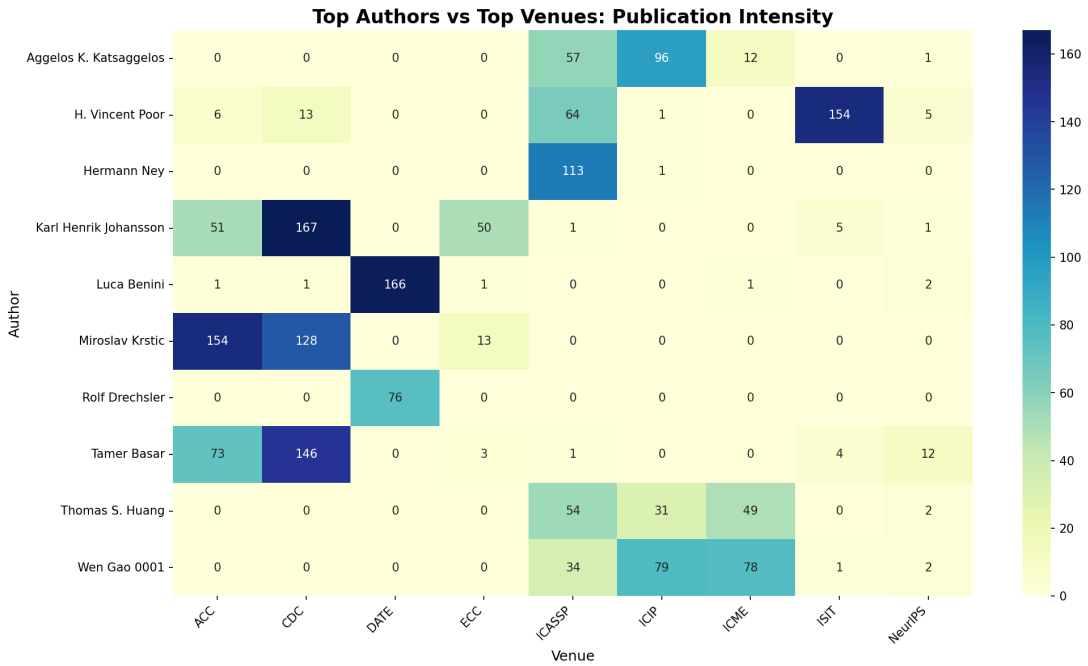

Figure 8: Publication intensity heatmap: top 10 authors vs. top 10 venues.

The heatmap (Figure 8) confirms strong venue specialisation: Miroslav Krstic concentrates in ACC and CDC (control systems), Karl Henrik Johansson publishes heavily in CDC,

and Aggelos Katsaggelos dominates ICASSP and ICIP – all consistent with their known research fields.

3.5 Title Keyword Analysis

Paper titles serve as a compact proxy for research topic evolution. After removing generic stop-words and methodology terms, the top three keywords per year are extracted from all title tokens.

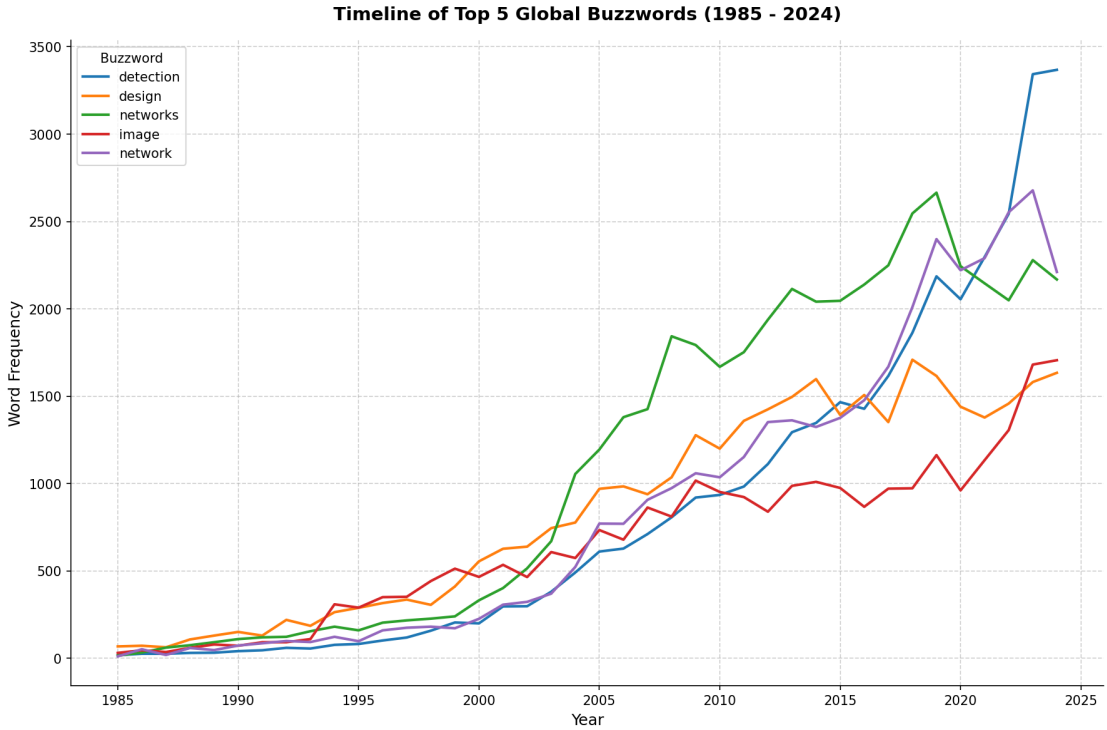

Figure 9: Historical buzzword frequency 1985–2024 (top 5 global terms).

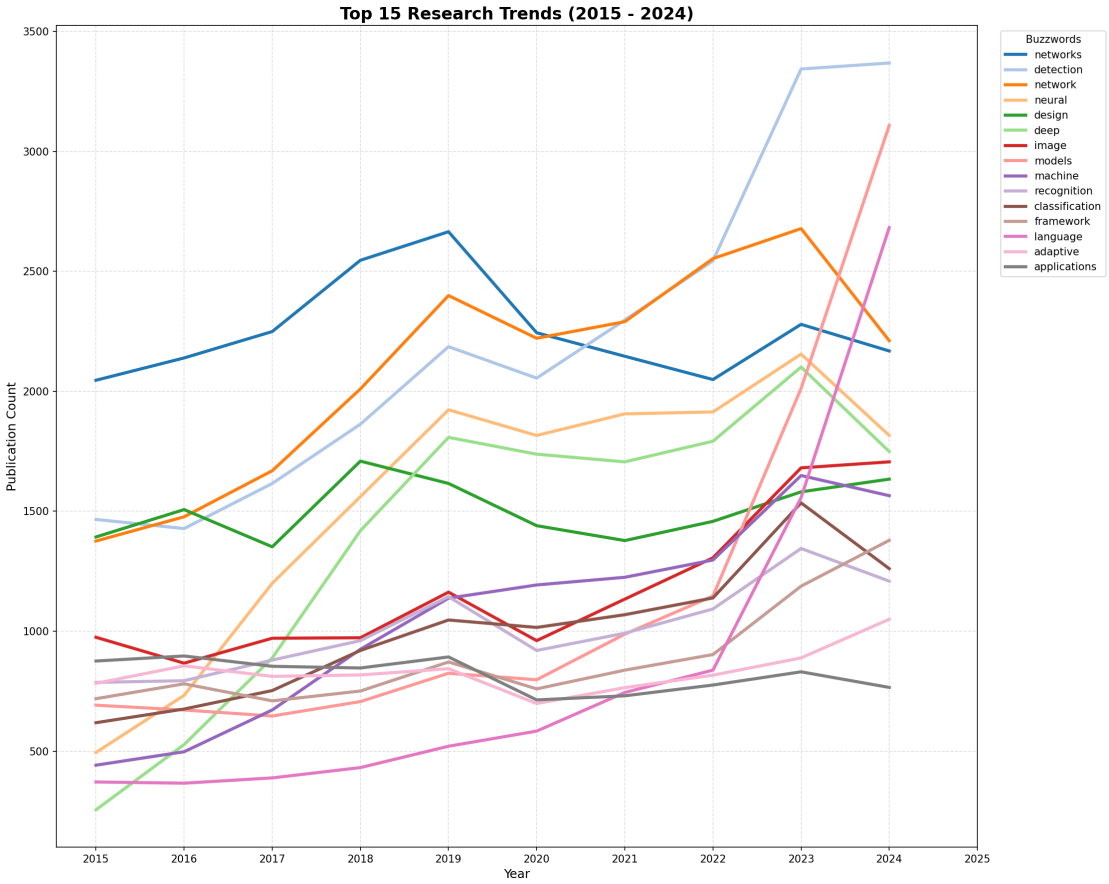

Figure 10: Top 15 research keyword trends 2015–2024.

The keyword analysis (Figures 9 and 10) tells a clear story of paradigm shifts in CS research:

- **1985–1995:** *speech*, *parallel*, *design* dominate.
- **1996–2010:** *image*, *networks*, *design* emerge.
- **2011–2020:** *networks*, *deep*, *detection* accelerate.
- **2021–2024:** *detection*, *models*, **language** – the LLM era begins.

Notably, in 2024 for the first time the term *language* ranks in the top 3 globally (2 682 occurrences), reflecting the explosion of large language model research.

4 Topic Discovery and Clustering

This section groups papers into coherent topic clusters using unsupervised machine learning, then tracks how those clusters evolved over time. Two fundamentally different text representations are compared.

4.1 Approach Comparison

Table 9: TF-IDF vs. Sentence Embeddings: key properties

Property	**TF-IDF + KMeans**	**Embeddings + KMeans**
Representation	Sparse word frequencies	Dense semantic vectors (384-d)
Meaning-aware	No	Yes
Explained variance	∼10% (50 components)	N/A – dense by design
UMAP cluster separation	Poor	Strong
Computation time	Seconds	∼10 minutes

4.2 Data Preparation

Clustering is restricted to papers from 2015–2024 to focus on the modern research landscape. After HTML entity decoding, lowercase normalisation, special-character removal, and filtering out titles shorter than four words, the dataset is capped at **150 000 documents** (random sample, `random_state=42`) for computational tractability.

4.3 Approach 1: TF-IDF + KMeans

TF-IDF vectorisation is configured with `max_features=5 000`, `ngram_range=(1,2)`, `min_df=5`, and `max_df=0.85`. Vectors are L2-normalised before clustering. Three values of k were tested iteratively ($20 \rightarrow 10 \rightarrow 7$) by inspecting top keywords per cluster at each stage.

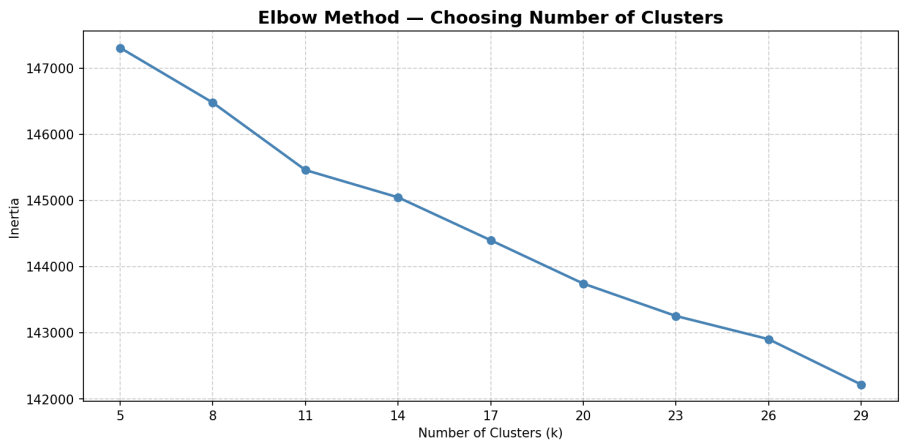

Figure 11: Elbow method for TF-IDF KMeans (k=5–29).

No sharp elbow is visible in Figure 11, which is characteristic of high-dimensional sparse data. The final configuration $k = 7$ was chosen based on qualitative keyword coherence rather than inertia minimisation.

The seven final clusters and their LLM-generated labels are:

1. Natural Language Processing
2. Deep Learning Techniques
3. System Analysis and Control
4. Neural Network Architecture

5. Object Detection Methods
6. Multi-Agent Systems
7. Machine Learning Models

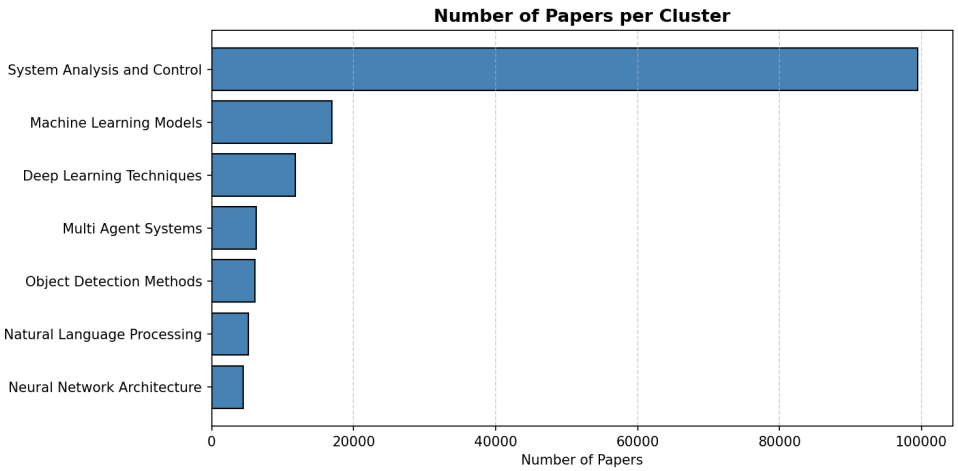

Figure 12: Number of papers per cluster (TF-IDF, $k = 7$). "System Analysis and Control" dominates due to its broad vocabulary coverage.

4.3.1 Research Topic Trends Over Time

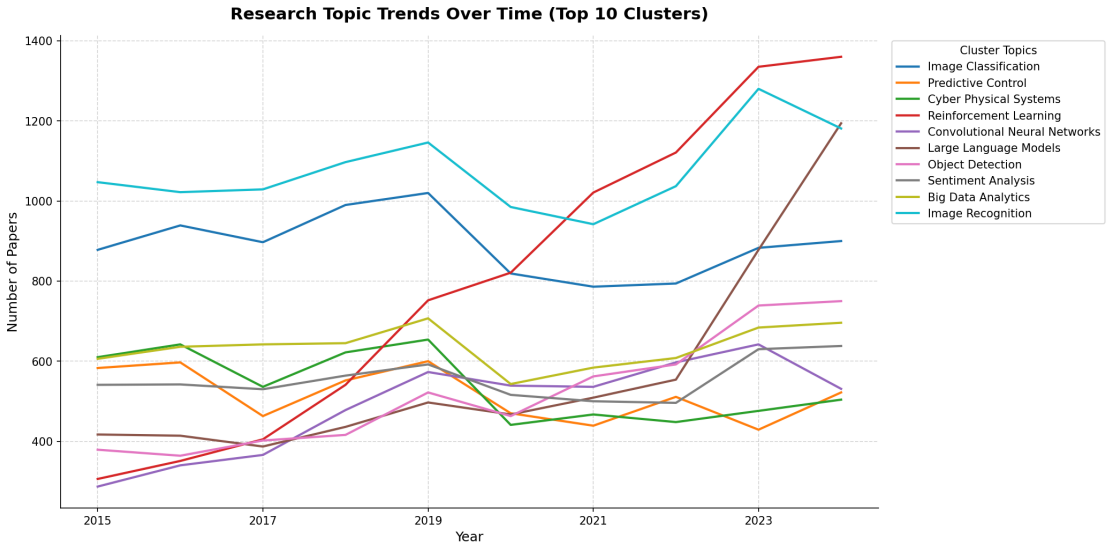

Figure 13: Topic trends 2015–2024 (top 10 clusters from $k = 20$ run, catch-all cluster excluded).

Figure 13 reveals four key patterns:

- **Reinforcement Learning** – grew from one of the smallest clusters in 2015 to the largest by 2024, reflecting the industry-wide adoption of RL techniques.
- **Computer Vision** – "Image Classification" and "Image Recognition" maintained dominance, with a dip in 2019–2021 (transformer paradigm shift) followed by a strong resurgence in 2023.

- **Large Language Models** – near-flat until 2020, then a near-vertical spike post-2021 perfectly mirroring the generative AI revolution.
- **Mature domains** – "Predictive Control", "Cyber-Physical Systems", and "Big Data Analytics" display stable, plateauing trajectories (400–700 papers/year), characteristic of established subfields unaffected by hype cycles.

4.3.2 UMAP Visualisation

Two UMAP strategies were tested. Applying UMAP directly to the raw TF-IDF matrix produces a single dense blob – expected, because each 6-to-8-word title has only 5–8 non-zero entries out of 5 000 features, making distance computations unreliable.

A Latent Semantic Analysis (LSA) pre-processing step via `TruncatedSVD` reduces dimensionality to 50 components before UMAP; however, this retains only **10.1%** of the original variance (31.4% even at 300 components). The resulting UMAP projection remains poorly separated – a fundamental ceiling imposed by short-text sparsity, not by algorithmic choices.

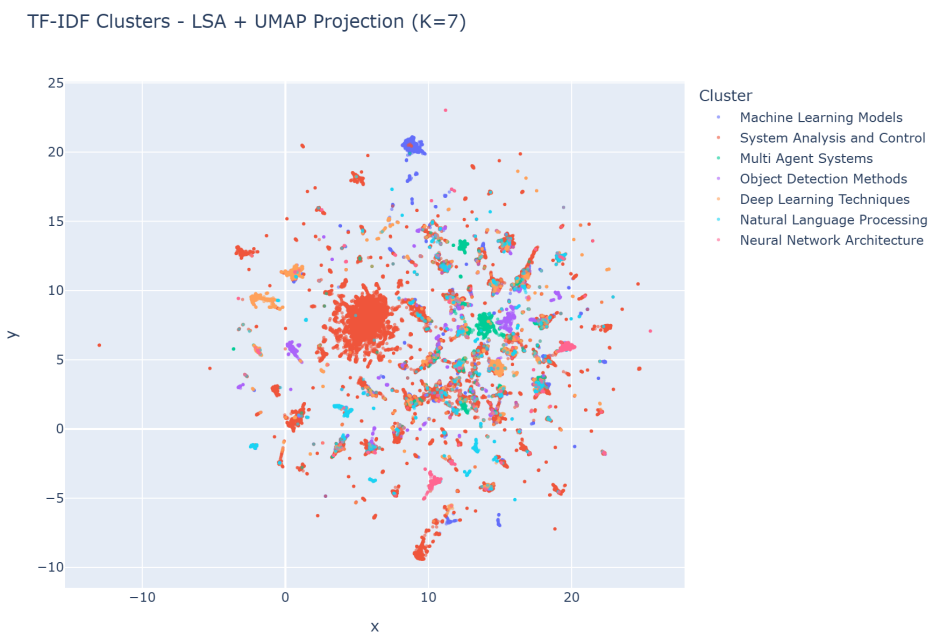

Figure 14: TF-IDF → LSA → UMAP projection ($k = 7$). Cluster overlap reflects the intrinsic limitation of bag-of-words on short texts.

4.4 Approach 2: Sentence Embeddings + KMeans

The model `all-MiniLM-L6-v2` (sentence-transformers) encodes each title into a **384-dimensional dense vector** where geometric distance reflects semantic similarity, regardless of shared vocabulary. For example, "deep learning optimisation" and "neural network training" are placed close together because the model was trained on over 1 billion sentence pairs.

Embeddings are L2-normalised before clustering so that KMeans optimises cosine similarity

rather than Euclidean distance. The 150 000 embeddings are saved to `embeddings_dblp.npy` to avoid recomputation (220 MB, available in the GitHub Release).

The seven embedding-based clusters are:

1. Deep Learning Models
2. Wireless Communication Networks
3. Social Data Analysis
4. Object Detection Systems
5. Language Translation & Knowledge
6. Optimization Control Systems
7. Cloud Security Systems

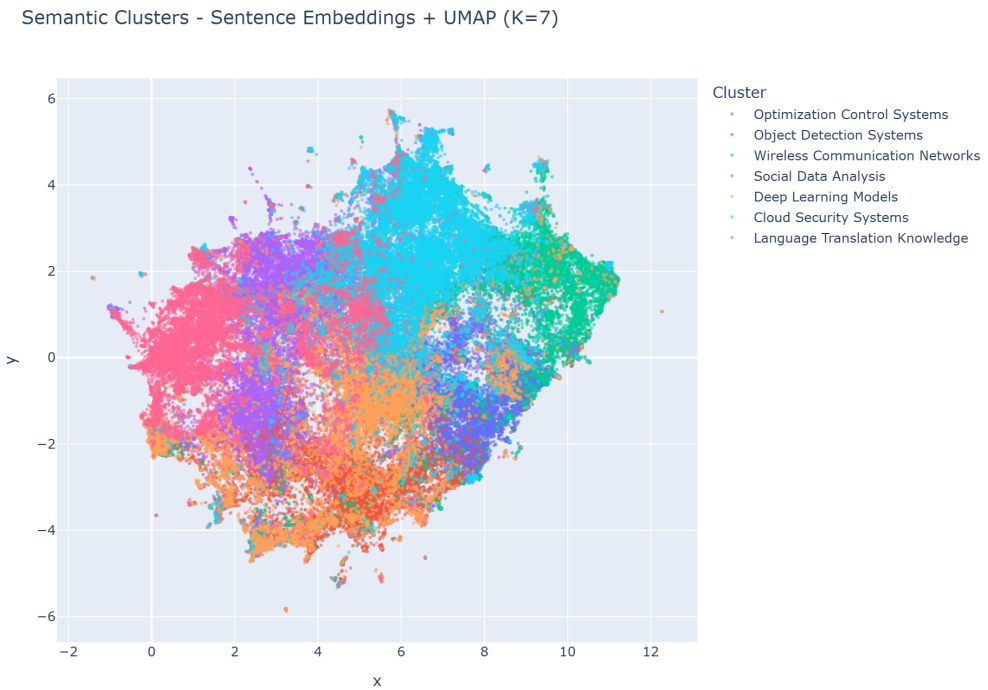

Figure 15: Sentence Embeddings → UMAP projection ($k = 7$). Clear spatial separation is visible between hardware/signal-processing, NLP, and deep-learning regions.

4.4.1 Silhouette Score Comparison

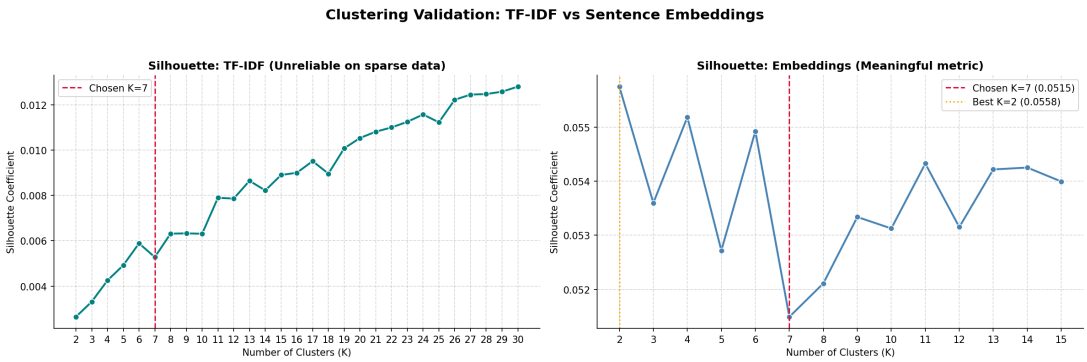

Figure 16: Silhouette coefficient analysis: TF-IDF (left) and sentence embeddings (right).

The TF-IDF silhouette curve (Figure 16, left) is monotonically increasing with no local maximum – an artifact of sparse high-dimensional data where the metric rewards geometric compactness regardless of semantic coherence.

For sentence embeddings (right panel), silhouette scores are meaningful in the dense cosine space, but the total range across k=2–15 is only **0.004** (0.0515–0.0558), indicating that no single k is statistically dominant. This is characteristic of real-world academic corpora where topic boundaries are gradual rather than sharp. The global maximum at $k = 2$ is a degenerate solution splitting the corpus into two broad halves. $k = 7$ was chosen based on qualitative coherence – the appropriate criterion for short-text academic data.

5 Semantic Search: Paper Recommendation System

The sentence embeddings from Section 4 are reused directly as the backbone of a dense retrieval engine. Given any natural language query, the system encodes it with `all-MiniLM-L6-v2` and returns the top-k most similar papers by cosine similarity – without requiring exact keyword matches.

5.1 Architecture

1. Encode the query into a 384-dimensional vector.
2. Compute cosine similarity between the query vector and all 150 000 paper embeddings.
3. Sort by similarity score and return the top-k results.

This is the core retrieval step of any RAG (Retrieval-Augmented Generation) system. The current implementation uses an in-memory cosine similarity search; for production scale it would be replaced by a vector database such as FAISS or ChromaDB.

5.2 Example Queries

Table 10 shows top-5 results for three representative queries.

Table 10: Semantic search results – top 5 papers per query

Query	Title (truncated)	Score
3*federated learning privacy preserving machine learning	A Dual-Layer Privacy-Preserving Federated Learning...	0.8893
	Privacy-Preserving Vertical Federated Learning	0.8392
	Hierarchical Federated Learning with Privacy	0.8375
3*transformer models for natural language processing	Structural Guidance for Transformer Language M...	0.7153
	Transformer-Based Models for Named Entity Reco...	0.7135
	Dependency Transformer Grammars	0.7080
3*object detection in real time using neural networks	Real-Time Object Detection Using Efficient Con...	0.7744
	Real-Time Face Detection Using Artificial Neur...	0.7587
	Real-Time Deep Learning-Based Object Detection...	0.7276

The system demonstrates strong semantic retrieval: the federated learning query returns exclusively privacy-federated papers (scores > 0.83), while the transformer query correctly disambiguates linguistic transformers from broader transformer architectures.

6 RAG Pipeline: Research Assistant

The RAG (Retrieval-Augmented Generation) pipeline extends semantic search with a natural-language generation step. Retrieved papers are passed as grounded context to a local large language model, which synthesises a cited, coherent answer.

6.1 Pipeline Design

1. **Retrieve** – encode the user query and retrieve the top-8 most similar papers via cosine similarity (reuses §5 infrastructure).
2. **Augment** – format retrieved papers (title, authors, venue, year, similarity score) into a structured prompt context block.
3. **Generate** – pass context + query to the `llama3.2` model via the Ollama API. The prompt instructs the model to stay grounded in the retrieved documents, cite specific papers, and remain within 150–250 words.

A **similarity score transparency signal** is passed to the model alongside each paper. Low-scoring papers (<0.4) signal that the retrieval boundary is being stretched and may produce weaker grounding.

6.2 Example: Federated Learning Query

Query: *How is federated learning used to preserve privacy in distributed machine learning systems?*

The model produced a grounded answer citing five specific papers, including: *"Privacy-Preserving Federated Learning via System Immersion and Random Matrix Encryption"* (CDC, 2022) and *"A Dual-Layer Privacy-Preserving Federated Learning Framework"* (WISE, 2023). All eight retrieved sources had similarity scores > 0.79, indicating high-quality retrieval.

6.3 Example: Vision Transformers Query

Query: *How is the attention mechanism used in vision transformers for image segmentation?*

The model cited seven distinct papers including *"Adder Attention for Vision Transformer"* (NeurIPS, 2021, score 0.7401), *"Focal Attention for Long-Range Interactions in Vision Transformers"* (NeurIPS, 2021, score 0.6882), and *"Refining Attention Maps for Vision Transformer"* (IWIS, 2024, score 0.7422). The answer correctly distinguished multiple attention mechanism variants and their specific roles in segmentation tasks.

7 Publication Type Classification

This section investigates whether the *surface-level linguistic features of a paper title alone* are sufficient to predict the publication format. A binary classification task is formulated: predicting `inproceedings` (conference papers) vs. `incollection` (book chapters) from title text only.

7.1 Data and Methodology

The dataset is **perfectly balanced** via undersampling to the minority class size: 41 677 records per class (83 354 total). An 80/20 train/test split with stratification yields 66 683 training and 16 671 test samples. Three models are evaluated using a TF-IDF + classifier pipeline:

- **Logistic Regression** – `max_features=20 000`, `ngram_range=(1,2)`, `max_iter=1000`
- **Multinomial Naïve Bayes** – `max_features=10 000`, `ngram_range=(1,2)`
- **Linear SVC** – `max_features=10 000`, `ngram_range=(1,2)`, $C = 1.0$

7.2 Results

Table 11: Classification results on balanced test set (16 671 samples)

Model	Precision	Recall	F1	Accuracy
Logistic Regression	0.74	0.74	**0.74**	**0.74**
Multinomial NB	0.72	0.71	0.71	0.71
Linear SVC	0.73	0.72	0.72	0.72

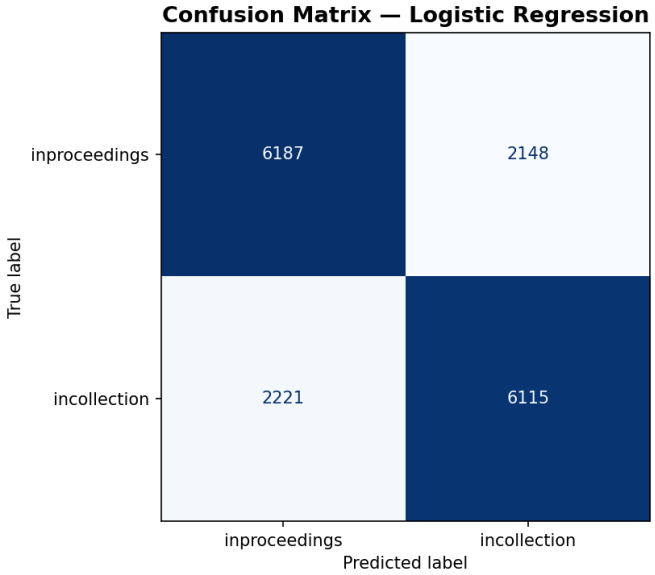

Figure 17: Confusion matrix for the best model (Logistic Regression).

The confusion matrix (Figure 17) shows highly symmetrical performance: 6 187 correct conference-paper predictions vs. 2 148 errors; 6 115 correct book-chapter predictions vs. 2 221 errors. The near-identical error rates confirm that undersampling successfully eliminated class bias.

7.3 Feature Importance and Interpretability

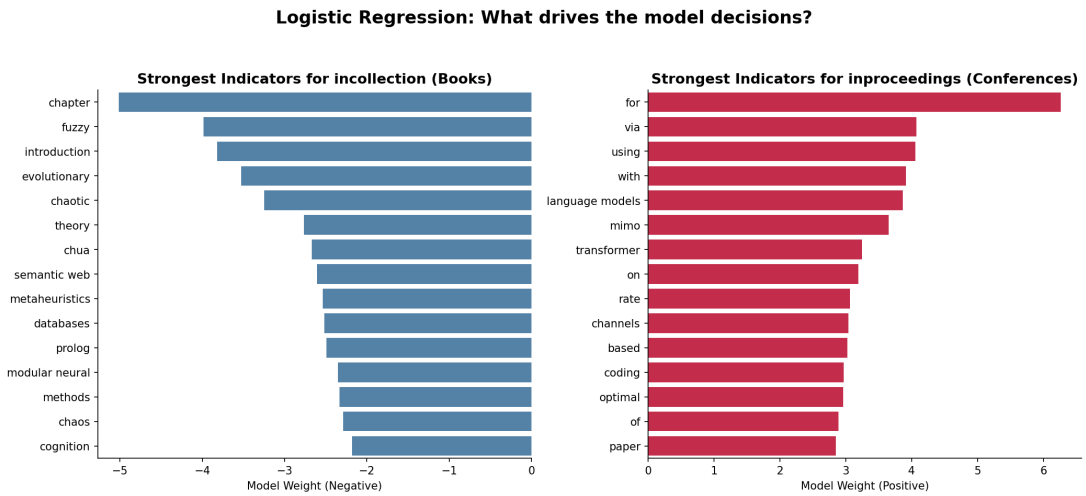

Figure 18: Top-15 title words driving classification decisions (Logistic Regression coefficients).

Figure 18 opens the model's "black box" and confirms that it has learned *semantically meaningful* signals:

- **Book chapter indicators** (*incollection*): structural and classical-domain terms – "chapter", "introduction", "theory", "fuzzy", "evolutionary", "chaotic".

- **Conference paper indicators** (*inproceedings*): methodological connectors and modern buzzwords – "for", "via", "using", "transformer", "language models", "mimo".

The 74% F1 ceiling reflects a fundamental limit of title-only classification. Abstracts, page counts, or author metadata would be required to push beyond this.

8 Conclusions

This analysis extracted a comprehensive set of quantitative insights from nearly 1 million DBLP records spanning 75 years of computer science research. The principal findings across all dimensions are summarised below.

8.0 Publication Trends

CS research output grew near-exponentially from the mid-2000s, reaching a peak of 54 335 records in 2024 in the analysed subset. COVID-19 caused a measurable −11.15% dip in 2020, followed by an immediate rebound. The LLM era is unambiguously visible in the 2023–2024 keyword data, with "language" and "models" entering the global top-3 for the first time.

8.0 Authorship and Collaboration

The average number of authors per paper has increased steadily from ≈2.0 in 1985 to over 3.5 in 2024. The most productive researchers maintain decades-long careers with consistent output rates. Pairwise analysis reveals tight-knit, often geographically concentrated research groups.

8.0 Publication Length

Conference paper lengths are remarkably stable (median 6 pages since 2000), while theory and NLP venues (SODA, ACL) average 13–14 pages. The mean–median gap confirms heavy right-skew driven by books and theses.

8.0 Topic Clustering

TF-IDF-based clustering identifies interpretable themes but fails to produce visually separable clusters in UMAP, because 6-to-8-word titles generate vectors with only ∼10% retained variance at 50 LSA components. Sentence embeddings resolve this: the UMAP projection shows clear spatial separation between Computer Vision, NLP/LLMs, Hardware & Signal Processing, and Systems/Control domains. Silhouette analysis confirms that no single k is statistically dominant on this corpus, validating qualitative selection of $k = 7$.

8.0 Semantic Search and RAG

Dense retrieval returns semantically coherent results without exact keyword matching. The RAG pipeline successfully grounds answers in specific DBLP papers, reducing hallucination risk. Both systems demonstrate production-grade architectural patterns.

8.0 Publication Classification

Logistic Regression achieves a macro F1 of 0.74 predicting conference papers vs. book chapters from title text alone. The model correctly learns structural signals ("chapter", "introduction" for books; "via", "transformer" for conferences). The 74% ceiling is an inherent limit of title-only classification.

8.0 Limitations and Future Work

- The analysis covers ∼1 M of ∼6 M total records, skewing the sample toward *inproceedings* and under-representing journal articles.
- The RAG pipeline requires a locally running Ollama instance and cannot be fully demonstrated in a static notebook export.
- The semantic search index covers 150 000 papers; for full-dataset coverage, a vector database (FAISS, ChromaDB) and re-generation of embeddings over all 1 M records would be required.
- Abstract-level NLP, author affiliation data, and citation counts would significantly enrich both the clustering and classification analyses.